



Department of Information Technology
Academic Year 2023 - 2024 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Conversion of NFA into DFA
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Collaborative Learning
- **Justification:**

In unit I, Conversion of NFA into DFA is an important topic in the unit. This activity helps the students to know the concepts and conversion from NFA to DFA.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The topic NFA to DFA taught in the class for 25 minutes and 5 minutes for summary of the topic.
 - The time allotted for the activity is 10 Minutes.
 - The students discussed and ask to discuss with their peers.
 - The students Aarthi S presented one example problem as shown in figure 1 and 2.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	1	2	1	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
	3	3	3	1	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of Theory of Computation for converting NFA to DFA.	The students will be able to identify the need for analysis of complex engineering problems	Able to design the process flow for converting NFA to DFA	Able to analyze the data for conversion	The students will be able to follow ethics for solving problems	The students able to function as a team for converting NFA to DFA	The students will be able to communicate one another	The students able to learn and implement the concept for Natural Language Processing	The students able to apply mathematical and computing skills.	The students can exhibit knowledge of languages in the field of Cloud Computing to develop IT solutions	The students able to exhibit knowledge in Languages to develop a quality product.

- Images / Screenshot of the practice:



Figure 1: Presented by Aarthi S and Vijayalakshmi S

1) Show that $7+13+19+\dots+(6n+1)=n(3n+4)$ for all n by mathematical induction

$$P(n) = 7+13+19+\dots+(6n+1) = n(3n+4)$$

$$n=1$$

$$L.H.S = [6(1)+1] = 7$$

$$R.H.S = 1(3(1)+4) = 7$$

$$P(1) \text{ is true}$$

Assume $P(k)$

$$7+13+19+\dots+(6k+1) = k(3k+4)$$

$$P(k) \text{ is true}$$

$P(k+1)$

$$= 7+13+19+\dots+[6(k+1)+1] = (k+1)[3(k+1)+4]$$

$$= (6k+1) + (6k+7) = (k+1)(3k+7)$$

$$k(3k+4) + (6k+7)$$

$$3k^2+4k+6k+7$$

$$3k^2+10k+7$$

$$(3k+7)(k+1)$$

2) Construct DFA equivalent to NFA $M = (\{p, q, r, s\}, \{a, b\}, p, \{s\}, \delta)$ where δ is defined in the following table

Sol:-

	δ	δ
p	$\{q, s\}$	$\{q, r\}$
q	$\{r\}$	$\{q, r\}$
r	$\{s\}$	$\{p\}$
s	-	$\{p\}$

Sol:-

$\delta(p, a) = \{q, s\}$	$\delta(\{q, s\}, a) = \{p, r, s\}$	$\delta(\{q, s\}, b) = \{q, r\}$
$\delta(p, b) = \{q, r\}$	$\delta(\{q, r\}, a) = \{p, q, r\}$	$\delta(\{q, r\}, b) = \{q, r\}$
$\delta(q, a) = \{r\}$	$\delta(\{r\}, a) = \{p\}$	$\delta(\{r\}, b) = \{p\}$
$\delta(r, a) = \{s\}$	$\delta(\{s\}, a) = \{p\}$	$\delta(\{s\}, b) = \{p\}$
$\delta(s, a) = \{p\}$	$\delta(\{p\}, a) = \{q, s\}$	$\delta(\{p\}, b) = \{q, r\}$

Transition table:-

	δ	δ
p	$\{q, s\}$	$\{q, r\}$
q	$\{r\}$	$\{q, r\}$
r	$\{s\}$	$\{p\}$
s	-	$\{p\}$
$\{p, q, r, s\}$	$\{q, s\}$	$\{p, q, r\}$
$\{p, q, r\}$	$\{r\}$	$\{p, q, r\}$
$\{p, q, s\}$	$\{r\}$	$\{p, q, r\}$
$\{p, r, s\}$	$\{s\}$	$\{p\}$
$\{q, r, s\}$	$\{r\}$	$\{q, r\}$

Figure 2 Solution given by Abinandhana

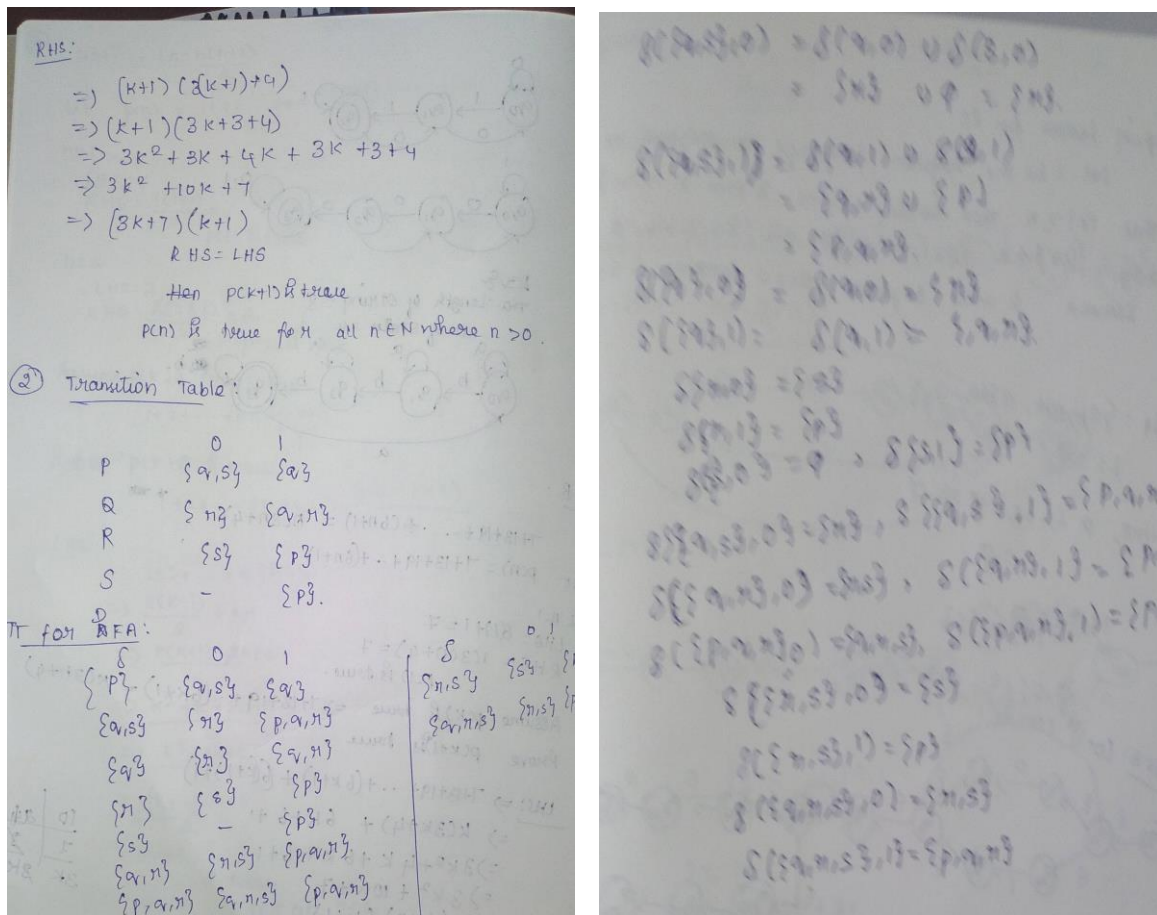


Figure 1 Solution given by Sobana S

Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- Students said that they were able to understand the concepts of NFA to DFA.
- The students said that the activity help to practice the concept of NFA to DFA.
- The students were shared their ideas for converting NFA to DFA.
- It makes the students to improve presentation skills.

❖ Benefit of the practice:

- Students can able to recall the methods for converting NFA to DFA.
- The students can share their ideas with their pairs.
- It helps the slow learners to understand the concepts.

❖ Challenges faced in implementation:

- The Activity planned for 10 minutes but it takes 20 minutes to complete the activity.
- Some students hesitate to present their solution.

References:

- ❖ <https://ohioline.osu.edu/>
- ❖ <https://www.nureva.com/blog/education/15-active-learning-activities-to-energize-your-next-college-class>
- ❖ <https://teaching.cornell.edu/resource/examples-collaborative-learning-or-group-work-activities>

Signature of Faculty Member

HOD